Notes

Combining polygenic scores

Selection index, three information routes,

and evidence in `multipgs`

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`https://github.com/bvilhjal/multipgs` · v0.3.3

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Abstract

A linear combination of existing polygenic scores is a selection index. Write Z for the vector of standardised component scores in a target population, $\rho_k = \mathrm{Corr}(Z_k, g_f)$ for the correlation of score k with the focal genetic value, and C for the correlation matrix of Z. The squared genetic-value correlation of a weight vector w is $(w^\top \rho)^2/(w^\top C w)$. When C is positive definite the optimum is $w_\star \propto C^{-1}\rho$. For the focal score and one auxiliary score the exact increment is $\Delta R^2 = r_G^2 R_k^2 (1 - R_f^2)^2/(1 - r_G^2 R_f^2 R_k^2)$. Absolute gain therefore falls as the square of the information still missing from the focal score.

The package implements three routes to an approximate index: penalised stacking from a phenotyped target cohort, the same Gaussian objective from score-space GWAS moments, and a training-free same-trait rule. They solve different information problems. In 240 held-out simulations where the genetic value lies in the score span, the individual stack beat the training-selected best single score in every replicate and closed 83.4–99.2% of the oracle gap. Exact individual moments recover the same coefficients (mean correlation 0.999624). A small LD reference and known discovery overlap inflate reported R^2 more than they degrade the fitted predictor. Decorrelation is exact under the intended ρ and collapses when that vector is replaced by a Daetwyler proxy.

These results establish numerical identities and controlled estimator behaviour. They do not establish clinical utility, ancestry portability, or real-world comparative superiority.

The intended reader already uses polygenic scores, GWAS summary statistics, and the distinction between a discovery sample and a target cohort. These notes do not re-derive single-trait PGS construction. They fix the combination estimand — many-trait score stacking, not a same-trait method ensemble — the selection-index algebra, the three information routes, and which simulation claims survive their design.

Contents

1 Introduction

A polygenic score (PGS) is a linear predictor of trait value built from a genome-wide association study (GWAS) [11, 21]. Its accuracy is bounded by discovery sample size, SNP heritability, and the effective number of independent effects [7, 11]. Many traits of interest have smaller discovery GWAS than genetically correlated phenotypes. The usual response is to borrow strength.

The same linear-index algebra is used on three different panels. Table 1 names them.

Family (a) combines *before* scoring: a multivariate model reweights variant-level summaries, as in MTAG or wMT-SBLUP [18, 17]. Family (b) builds one score per source and combines the scores. That family still splits. One branch ensembles several constructions of the *same* trait — methods, ancestries, or GWAS — as in metaGRS, PRSmix, PUMAS-ensemble, and MIXPRS [15, 27, 25, 28]. The other branch stacks scores of *many traits*, as in Krapohl, Albiñana, and PRSmix+ [16, 26, 27, 23]. Family (c) derives the combination weights from metadata rather than from a target phenotype [17, 1, 2].

MIXPRS combines JointPRS-auto and SDPRX for one trait across ancestries. Its weights are non-negative least squares on a data-fission split of the target GWAS; SNP pruning plus an identity residual is used so that a mismatched LD reference does not correlate the pseudo-training and pseudo-tuning residuals [28]. That is a same-trait method ensemble. It is not multi-PGS. The MIXPRS discussion lists PRS for related traits as future work; that is the problem this package addresses.

Table 1: Neighbouring combination methods, distinguished by the object in the linear index. `multipgs` implements the last two rows.

Object	**Examples**	**This package**
Variant effects of several traits, before scoring	MTAG, wMT-SBLUP	Not implemented
Same-trait methods or ancestries	MIXPRS, PUMAS-ensemble, metaGRS, PRSmix	`meta_pgs` is the metadata-weight case only; MIXPRS itself is not implemented
Scores of many traits	Albiñana, Krapohl, PRSmix+	`multi_pgs_fit`, `multi_pgs_sumstats`

The published Multi-PGS analysis trained standardised scores with unpenalised covariates and reported fivefold out-of-sample assessment [26]. `multipgs` implements that broad estimator, but it is not a reproduction of the paper's procedure. The package standardises inside training folds, selects and averages coefficients by cross-model selection and averaging (CMSA) [20], adds a nested fallback gate, supports a score-space summary-statistic objective, and reports an unnormalised incremental R^2. Those differences are identifying, not cosmetic.

The rest of the note is the estimand (§2), the selection-index model (§3), the three routes (§4), deployment and metrics (§5), the simulation evidence (§6), the unit-test contracts (§7), and the interpretation limits (§8).

2 Estimand

Write S_{ik} for the raw allele-weighted score of individual i on component $k = 1, \ldots, K$, and write

$$Z_{ik} = \frac{S_{ik} - \mu_k}{\sigma_k}, \qquad \beta_k = \frac{\theta_k}{\sigma_k}, \tag{1}$$

for the training-standardised score and the corresponding raw-scale coefficient. Penalisation is defined on θ. Confusing these coordinates changes the meaning of every penalty factor. Notation used throughout is collected in Table 2.

The package does not discover variants or estimate GWAS effects. It starts from component weights that already exist. The object it returns is a single deployable variant-weight vector

$$\widetilde{w}_j = \sum_{k=1}^{K} w_{jk}\beta_k, \tag{2}$$

where w_{jk} is the component weight of variant j in score k. Assessment of $\widetilde{w}$ must use data untouched by component discovery, combination fitting, and tuning. That is the scientific claim. Everything else is a route to β.

Three estimands share that deployable object and are otherwise distinct.

Learned combination. Given a phenotyped target sample, or given score-space moments that stand in for one, the target is a penalised predictive coefficient θ for the observed phenotype y in the span of the supplied scores. The coefficients are not genetic correlations, not mediation weights, and not a feature-selection discovery.

Same-trait index. When every component estimates the same genetic value, the target is the selection-index direction $w \propto C^{-1}\rho$ under a stated model for ρ. Metadata enter only through that model.

What is not an estimand here. Absolute risk, a calibrated probability, an ancestry-adjusted mean, or a causal decomposition of the panel. A method-by-ancestry ensemble of one trait, as in MIXPRS [28], is a neighbouring estimand and is not implemented. A high ranking metric is not any of those things.

Two quantities are both called “the accuracy of a PGS”. Write $R_k = \mathrm{Corr}(Z_k, g_k)$ for genetic-value accuracy and $h_k R_k = \mathrm{Corr}(Z_k, y_k)$ for phenotypic accuracy. The package function `daetwyler_r2` returns phenotypic R^2, $h_k^2 R_k^2$. Taking its square root therefore gives $h_k R_k$, not R_k. Within one trait h cancels from any weight direction; across a panel of different traits it does not.

3 Selection index

A useful working model writes each standardised component score as

$$Z_k = R_k g_k + \sqrt{1 - R_k^2}\,\varepsilon_k, \tag{3}$$

Table 2: Notation used consistently in this report.

Symbol	Meaning
n, K	number of target individuals and number of component scores
S, Z	raw and training-standardised score matrices
θ, β	standardised-score and raw-score coefficients, related by (1)
g_k, g_f	standardised additive genetic values for component trait k and focal trait f
R_k	$\mathrm{Corr}(Z_k, g_k)$, genetic-value accuracy of score k for its own trait
$r_G(k, \ell)$	target-population genetic correlation between traits k and ℓ
ρ	target-population vector with $\rho_k = \mathrm{Corr}(Z_k, g_f)$
C	correlation matrix of the standardised component scores in the target population
N_k, M	effective GWAS sample size and effective number of independent effects or segments
π, p_s	population prevalence and sample case fraction for a binary trait

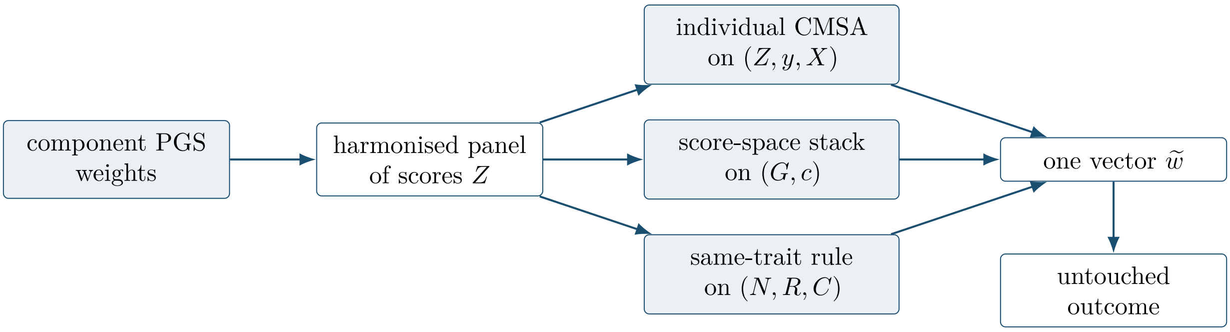

Figure 1: One deployable score, three information routes. The routes are not interchangeable. Assessment data must be untouched by component discovery, combination fitting, and tuning.

where $\mathrm{Var}(\varepsilon_k) = 1$, every ε_k is orthogonal to every genetic value in the model, and the residual errors may be correlated across scores. The last orthogonality condition is a mediation and transport assumption; the algebraic decomposition of Z_k with respect to g_k alone does not imply it.

Under independent residual errors,

$$\rho_k = R_k r_G(k, f), \qquad C_{k\ell} = \begin{cases} 1, & k = \ell, \\ R_k R_\ell r_G(k, \ell), & k \neq \ell. \end{cases} \tag{4}$$

Discovery sample overlap is one mechanism that can correlate the residuals. Shared pipelines, LD references, model fitting, and other common estimation artifacts can do so as well. Conversely, overlap need not induce a large covariance in every design.

For a weight vector w, the squared genetic-value correlation of the linear index is

$$R^2(w) = \frac{(w^\top \rho)^2}{w^\top C w}. \tag{5}$$

When C is positive definite, Cauchy–Schwarz gives

$$w_\star \propto C^{-1}\rho, \qquad R^2_{\max} = \rho^\top C^{-1} \rho. \tag{6}$$

This is the classical selection-index solution [1, 2]. The same algebra combines correlated forecasts [3]. A pseudoinverse or ridge is needed when C is singular. Neither repairs a misspecified ρ.

Closed-form gain from one auxiliary trait. For the focal score and one auxiliary score with genetic correlation r, the exact increment above the focal score is

$$\Delta R^2 = \frac{r^2 R_k^2 (1 - R_f^2)^2}{1 - r^2 R_f^2 R_k^2}. \tag{7}$$

Three factors are visible: squared genetic correlation, squared auxiliary accuracy, and the squared information still missing from the focal score. A large auxiliary GWAS cannot compensate for genetic irrelevance, and a well-powered focal score leaves less room to gain.

The Daetwyler reference model relates genetic-value accuracy to GWAS power [7],

$$R_k^2 = \frac{x_k}{1+x_k}, \qquad x_k = \frac{N_k h_k^2}{M}. \tag{8}$$

It is an upper-reference model under strong architecture and population assumptions, not a per-score truth oracle. The numerator of (7) then carries $(1+x_f)^{-2}$: the absolute gain falls as the square of the focal GWAS's own power. Table 3 records the arithmetic at $r_G = 0.5$ and auxiliary $R_k^2 = 0.5$. Figure 2 plots the same equation against focal sample size. Both displays are equation evaluations, not empirical performance claims. The same asymmetry is long established in multivariate genomic prediction: gains concentrate on the underpowered trait when a correlated, better-measured trait is available [9]. It is also the shape of the published Multi-PGS result, whose largest relative increase landed on ADHD [26].

Table 3: Exact two-score genetic-value increment from (7) under (8), with $r_G = 0.5$, auxiliary $R_k^2 = 0.5$, $h_f^2 = 0.2$, and $M = 20{,}000$, so $x_f = N_f/100{,}000$. Phenotypic increments are these values times h_f^2. The relative column is unaffected by that scaling.

N_f	x_f	R_f^2 alone	ΔR^2	relative gain
10,000	0.1	0.091	0.104	+115%
100,000	1	0.500	0.033	+6.7%
400,000	4	0.800	0.006	+0.7%
1,000,000	10	0.909	0.001	+0.1%

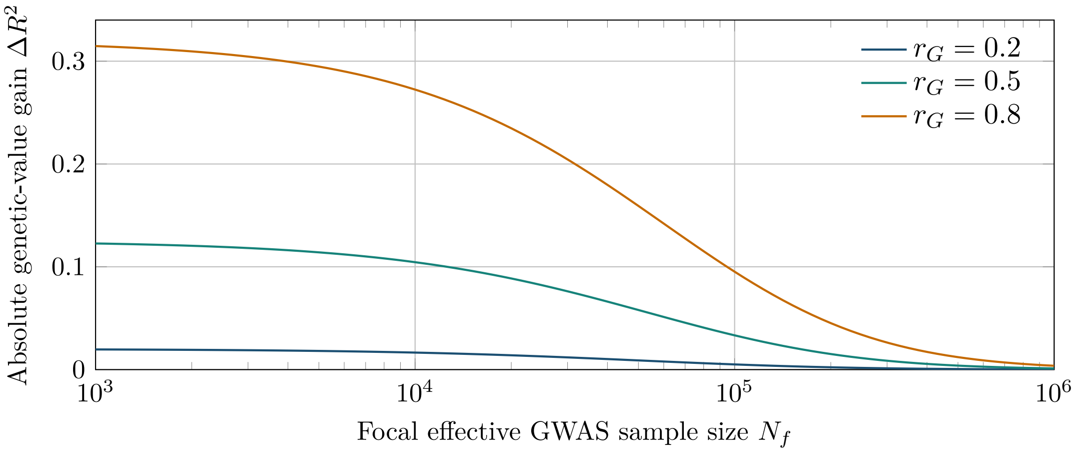

Figure 2: Analytical two-score gain from (7). The illustration fixes $h_f^2 = 0.2$, $M = 20{,}000$, and auxiliary $R_k^2 = 0.5$, so $x_f = N_f/100{,}000$. It is an equation plot, not an empirical performance claim.

4 Three information routes

Table 4 states the information contract. The choice among routes is determined by the data that exist, not by a single ranking of methods.

Table 4: Information and validation contract for the three combination routes.

Route	Learns from	Selection / tuning	Valid assessment
`multi_pgs_fit`	Individual score matrix, target phenotype, optional covariates	Inner CMSA choices inside outer folds; final coefficients average fold-selected fits; nested heuristic may return the baseline	Independent individuals who trained neither component scores nor the combination
`multi_pgs_sumstats`	Target-GWAS cross-moments plus external LD score Gram	Same moments, an independent second GWAS, or PUMAS-style pseudo-splits	A third untouched GWAS or independent individual-level cohort; tuning performance is not assessment
`meta_pgs`	Same-trait scores plus effective sample sizes or expected accuracies; target score correlations for decorrelation	No fitted tuning; ridge only stabilises inversion	Independent target outcomes; simple metadata-derived weights remain hypotheses until tested

4.1 Individual-level penalised stacking

Within each training fold, score columns are imputed and standardised from that fold alone. With unpenalised covariates X, write $\eta_i = b_0 + X_i^\mathsf{T}\gamma + Z_i^\mathsf{T}\theta$. The Gaussian or binomial elastic-net objective is

$$\min_{b_0,\gamma,\theta} \mathcal{Q}(y,\eta) + \lambda \sum_{k=1}^{K} p_k \Big\{\alpha|\theta_k| + \tfrac{1-\alpha}{2}\theta_k^2\Big\}, \tag{9}$$

where

$$\mathcal{Q}(y,\eta) = \begin{cases} (2n)^{-1}\sum_i (y_i-\eta_i)^2, & \text{Gaussian,} \\ -n^{-1}\sum_i\{y_i\eta_i - \log(1+\exp\eta_i)\}, & \text{binomial.} \end{cases} \tag{10}$$

Covariates receive penalty factor zero. The Gaussian path uses covariance updates; the binomial path uses IRLS with coordinate descent [8, 5]. For a standardised Gaussian column with current partial score $z_k^\star$, the coordinate update is

$$\theta_k \leftarrow \frac{\operatorname{sign}(z_k^\star)(|z_k^\star| - \lambda\alpha p_k)_+}{G_{kk} + \lambda(1-\alpha)p_k}. \tag{11}$$

In the Gaussian case, unpenalised covariate columns are equivalent to residualising first (Frisch–Waugh–Lovell). The equivalence does not hold for the binomial loss, where the IRLS weights depend on the current fit.

CMSA fits a path inside folds, selects a fold-specific model, and averages the selected coefficient vectors [20]. Averaging is a variance reduction for an unstable selector: with correlated scores and an ℓ_1 penalty, which of a set of near-duplicates is picked is close to arbitrary. The averaged support is the union of fold supports. Consequently `selected()` reports predictive weights, not causal traits, unique genetic contributors, or stable feature-selection discoveries.

The package adds a nested outer-fold decision rule that falls back to the unpenalised baseline when the apparent increment is insufficient. Choosing a tuning parameter and assessing performance are different tasks [4, 6]. An ordinary CMSA validation fold cannot also assess the stack: its phenotype selected that fold's (α, λ). The nested estimate cv-R^2 is a predictive loss gain of the inner stack over the explicit unpenalised baseline on untouched outer rows. It is not the OLS-recalibrated incremental R^2 of §5. The one-standard-error fallback fed by this estimate is a deployment heuristic, not a hypothesis test. In the committed null benchmark, pure noise passed in 11 of 60 checked runs.

4.2 Summary-statistic score-space stacking

Let W_{LD} contain component weights on the LD source's standardised-genotype scale, D be its LD correlation matrix, and W_{GWAS} contain the same component scores on the GWAS source's standardised-genotype scale. The sufficient moments are

$$G_{\mathrm{raw}} = W_{\mathrm{LD}}^{\mathsf{T}} D W_{\mathrm{LD}}, \qquad c_{\mathrm{raw}} = W_{\mathrm{GWAS}}^{\mathsf{T}} z. \tag{12}$$

Writing $A = \mathrm{diag}(\sqrt{\mathrm{diag}(G_{\mathrm{raw}})})$, the fitted coordinates use

$$G = A^{-1} G_{\mathrm{raw}} A^{-1}, \qquad c = A^{-1} c_{\mathrm{raw}}. \tag{13}$$

This is the same Gaussian objective as (9), but in the span of the supplied scores. At $\alpha = 1$, the lasso selects entire component scores. It is inspired by the SNP-level lassosum quadratic [13], but it is not SNP-level lassosum: the fitted variant effects cannot leave that span.

For a fixed coefficient vector, summary tuning evaluates

$$\mathrm{MSE}(\theta) = \mathrm{var}(y) - 2\theta^{\mathsf{T}} c + \theta^{\mathsf{T}} G \theta, \qquad \widetilde{R}^2(\theta) = \frac{(\theta^{\mathsf{T}} c)^2}{\mathrm{var}(y)\, \theta^{\mathsf{T}} G \theta}. \tag{14}$$

MSE, not squared correlation, selects the path because R^2 is blind to a reversed predictor. An independent second GWAS can tune the path; a third is still needed for assessment. PUMAS-style pseudo-splitting uses a joint-Gaussian/CLT covariance approximation [24]. In the committed calibration, relative covariance error is 3.77% for Gaussian and 10.99% for binary simulations. That quantifies the approximation. It does not validate pseudotuning as assessment. Every scalar and pseudo-split path carries coordinate-descent status; iteration-exhausted candidates are excluded rather than silently selected.

When D and z come from the same individuals, G and c are exact sample moments. With an external LD reference they are plug-in estimates. Sampling noise can put c outside the range of G, or make the plug-in R^2 exceed one. Those are diagnostics, not automatic proof of bad data.

4.3 Training-free same-trait combination

The simple rules standardise component scores and use

$$w_k \propto \sqrt{N_{\mathrm{eff},k}} \quad \text{or} \quad w_k \propto \sqrt{\widehat{R}_k^2}. \tag{15}$$

The first is a power-limited approximation sourced from pipeline code, not from the Hansen et al. preprint [29]. Under independent estimation errors, the exact same-trait GLS direction is

$$w_k \propto \frac{R_k}{1 - R_k^2}, \tag{16}$$

so direct accuracy weighting is accurate only as every R_k approaches zero. Accuracy saturates; the optimal weight does not. Weighting by accuracy therefore under-weights a well-powered GWAS more than weighting by sample size does.

The decorrelated rule is

$$w_\delta \propto (C + \delta I)^{-1} \rho. \tag{17}$$

Its algebra is exact under (6), but small error in ρ is amplified along low-eigenvalue directions of C. Ridge reduces that amplification only by moving the result back toward an undecorrelated weighted sum. The implemented ρ is a nonnegative magnitude. Every component must first be oriented to the same positive target direction; the rule cannot represent a genuinely negative target correlation. If only Daetwyler proxies are available, the checked real CAD diagnostic supports `expected_r2`, not `decorrelated`.

The same-trait assumption $r_G(k, \ell) \approx 1$ is assumed, not known. Nominally identical phenotype GWAS can have $r_G < 1$ when definitions or ascertainment differ. Applied to genuinely different traits the assumption is wrong, and a learned combination is the right tool.

5 Deployment and evaluation

Catalog scoring files are aligned by `panel_from_catalog`. Summary-statistic components are built by `panel_from_sumstats` through ldpred3 [22]. Identifier order, genome build, effect orientation, and source-specific dosage standard deviation are part of the statistical contract, not file-format trivia.

Write component k's term as $c_{jk}(g_j - \mu_{jk})$, with $c_{jk} = w_{jk}/\sigma_{jk}$ for a standardized score and $c_{jk} = w_{jk}$ for a raw score. Define $C_j = \sum_k \beta_k c_{jk}$ and $M_j = \sum_k \beta_k c_{jk}\mu_{jk}$. The exact one-row frozen representation has

$$\mu_j^\star = \frac{M_j}{C_j}, \qquad \widetilde{w}_j = \sigma_j^\star C_j. \tag{18}$$

Raw allele-count components use the target mean and leave one removable intercept. If $C_j = 0, M_j \neq 0$, or $\mu_j^\star \notin [0, 2]$, no valid single LDpred3 row can preserve both observed dosage and reference-mean missing imputation, so deployment is refused rather than silently rescaled.

The frozen scoring scale should be carried to new cohorts. Re-standardising each cohort independently can change coefficients, calibration, and even the meaning of a fixed threshold.

With covariates, the package reports the unnormalised increment

$$R^2_{\text{incremental}} = R^2_{\text{covariates+score}} - R^2_{\text{covariates}}. \tag{19}$$

The Albiñana paper instead divides this difference by $1 - R^2_{\text{covariates}}$ [26]. Absolute values are therefore not identical even when relative comparisons share covariates. Ancestry principal components belong in that covariate set. Without them a score can predict ancestry rather than disease [12].

For binary outcomes, with population prevalence π, sample case fraction p_s, threshold $t = \Phi^{-1}(1-\pi)$, density $\varphi(t)$, and $i = \varphi(t)/\pi$, the implemented Lee transformation is

$$C_{\text{LT}} = \frac{[\pi(1-\pi)]^2}{\varphi(t)^2 p_s(1-p_s)}, \quad \vartheta = \frac{i(p_s-\pi)}{1-\pi}\left\{\frac{i(p_s-\pi)}{1-\pi} - t\right\}, \quad R_L^2 = \frac{C_{\text{LT}} R_O^2}{1 + C_{\text{LT}}\vartheta R_O^2}. \tag{20}$$

This metric depends on a stated prevalence assumption [10]. Observed-scale R^2 is not comparable across ascertainment schemes.

Squared correlation hides sign reversal. Report signed correlation, or at least verify its sign, alongside R^2. Neither a weighted sum nor a high R^2 is a calibrated absolute risk.

6 Simulation evidence

The validation evidence is organised into four levels that are not pooled:

1. algebraic identities and numerical invariants;
2. unit and contract tests (§7);
3. replicated synthetic or real-LD simulations with checked raw rows;
4. descriptive real-data diagnostics without an untouched outcome truth.

A passing simulation establishes behaviour under that simulation. It does not establish biological transport or clinical benefit. Headline numbers below are derived from the committed CSVs named in each caption; Appendix A records the digest.

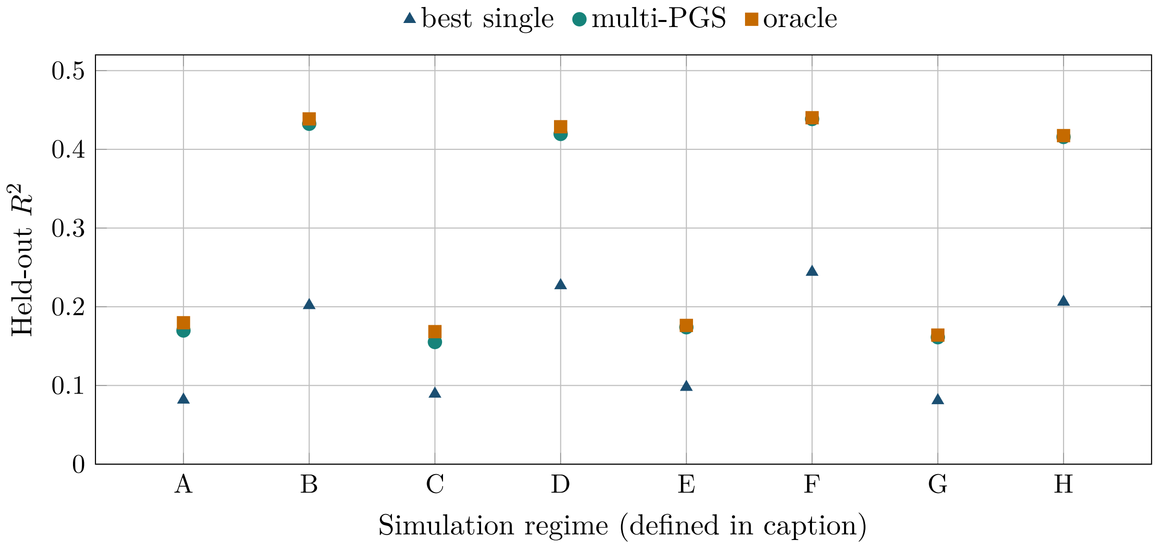

Figure 3: Mean held-out R^2 over 30 seeds from `benchmarks/results/fit_accuracy_summary.csv`. Regimes A–D use $n = 2000$, E–H use $n = 10{,}000$; A,B,E,F use $K = 50$, the others $K = 200$; A,C,E,G use $h^2 = 0.2$, the others $h^2 = 0.5$. The multi-PGS beats the training-selected best single score in 240 of 240 raw replicates. Because the simulated genetic value lies in the score span, this is an estimator sanity check, not evidence of clinical efficacy.

6.1 Individual stacking as an estimator

Figure 3 reports mean held-out R^2 over 30 seeds in each of eight regimes. Because the simulated genetic value lies in the score span, this is an estimator sanity check, not evidence of clinical efficacy.

Across the eight regimes, mean uplift over the selected single score ranges from 0.0660 to 0.2307 R^2. Based on regime means, the fitted stack closes 83.4%–99.2% of the gap between the selected single score and the oracle. The overall mean nested-CV minus held-out difference is -0.00230, slightly conservative in this simulation. Support recall is high (0.84–1.00 by regime) while precision is only 0.09–0.20: the averaged union support overstates sparsity, as the CMSA construction predicts.

6.2 Score-space identity and LD-reference sensitivity

With exact individual moments, the summary and individual standardised coefficients correlate 0.999858 on average, and all compared held-out R^2 values lie between 0.4973 and 0.4983. This supports the score-space identity under the benchmark's strong-signal, small-K design.

Figure 4 replaces the exact Gram by a real-LD reference of finite size. At reference $n = 50$, true fitted R^2 falls by 0.01795 relative to the infinite-reference result, while the own-reference estimate is inflated by 0.04446 (26.4% of true fitted R^2). Reporting bias is larger than the loss in the fitted predictor.

6.3 Overlap inflation

Figure 5 injects a known residual-error correlation. True predictive R^2 barely moves. Reported R^2 does. At overlap 0.10, reported-to-true R^2 ratios are 2.03 for multi-PGS and 4.89 for the single score; at 0.25, they are 3.98 and 14.73. Cross-validation inside the target cohort cannot remove discovery contamination shared by every fold [12].

Bivariate LDSC detected 16 of 16 non-null runs and 0 of 8 null runs in this 109,590-variant design. That does not imply universal LDSC power. It refutes the absolute claim that no

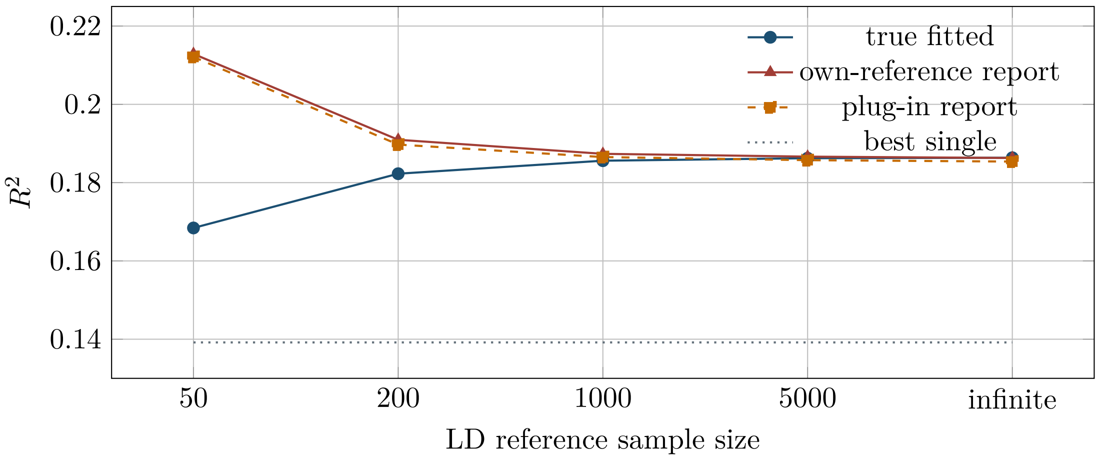

Figure 4: Real-LD simulation from `real_ld_simulation_summary.csv`. At reference $n = 50$, true fitted R^2 falls by 0.01795 relative to the infinite-reference result, while the own-reference estimate is inflated by 0.04446 (26.4% of true fitted R^2). Reporting bias is larger than the loss in the fitted predictor.

downstream diagnostic can ever flag overlap. Exclusion by design remains stronger than any post hoc test.

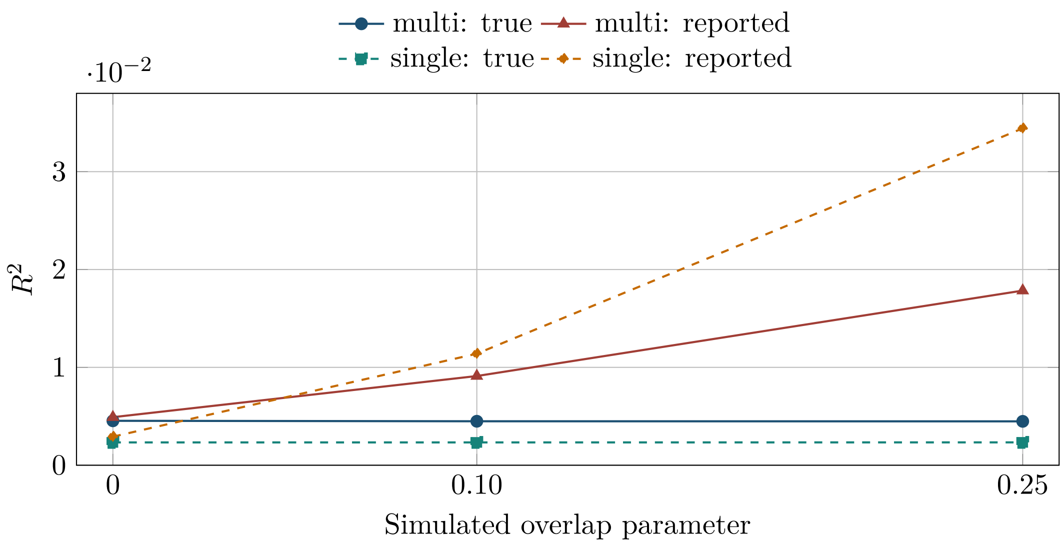

Figure 5: Known-overlap simulation from `overlap_inflation_summary.csv`. At overlap 0.10, reported-to-true R^2 ratios are 2.03 for multi-PGS and 4.89 for the single score; at 0.25, they are 3.98 and 14.73. Bivariate LDSC detected 16 of 16 non-null runs and 0 of 8 null runs in this 109,590-variant design.

6.4 Decorrelation and proxy ρ

Figure 6 contrasts two regimes. On the left, the simulation supplies the intended accuracy structure: the decorrelated rule holds its held-out R^2 as shared error grows, while the simple rules decline. On the right, a single checked 24-score CAD diagnostic replaces that ρ by Daetwyler or sample-size proxies. Decorrelation then collapses. The right panel is a contaminated regime-C score-space description, not external outcome truth. The direct expected-R^2 proxy is 0.17729, versus 0.002261 after decorrelation.

Hence: the algebra of (17) is not a licence to invert a poorly estimated ρ. A well-estimated C does not repair that.

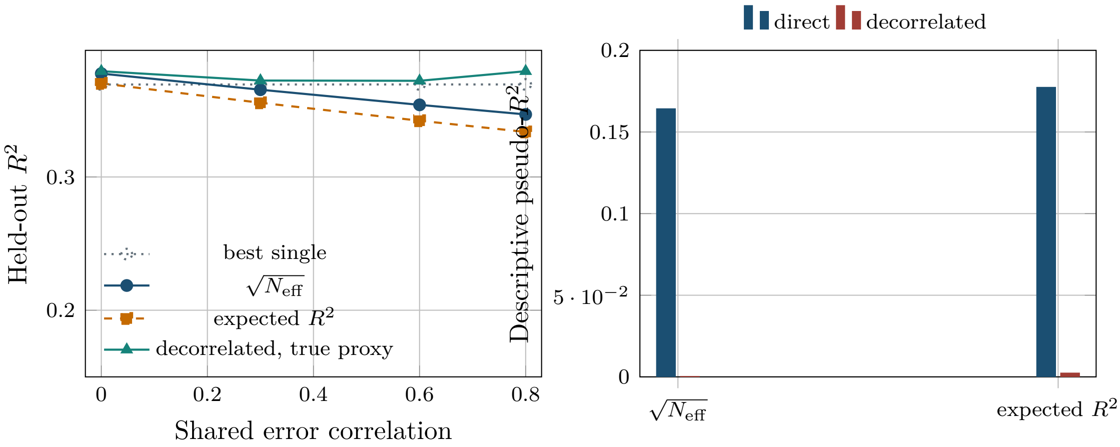

Figure 6: Decorrelation behaves well when the simulation supplies the intended accuracy structure (left, 30 seeds per point), but collapses with proxy accuracies in the single checked 24-score CAD diagnostic (right). The right panel is a contaminated regime-C score-space description, not external outcome truth. Sources: `meta_rules_summary.csv` and `real_meta_rules_summary.csv`. The direct expected-R^2 proxy is 0.17729, versus 0.002261 after decorrelation.

Table 5 restates the headlines by evidence level. The interpretations are part of the result.

Table 5: Headline validation evidence, separated by evidence level.

Artifact	**Result**	**Interpretation**
`fit_accuracy`	Multi-PGS wins 240/240 held-out simulations; uplift 0.0660–0.2307	Supports estimator sanity when truth lies in the score span; does not establish real-outcome or clinical superiority.
`null_gate`	Noise passes 11/60 runs overall	Shows the fallback is a decision heuristic, not a significance test.
`sumstat_vs_individual`	coefficient correlation 0.999858; R^2 0.4973–0.4983	Supports exact-moment Gaussian equivalence in a small-K, strong-signal design.
`sumstat_calibration`	PUMAS covariance relative error 3.77% Gaussian, 10.99% binary	Quantifies approximation error; it does not validate pseudotuning as assessment.
`real_ld_simulation`	reference $n = 50$ report inflation 0.04446	Demonstrates reference-induced optimism under real LD.
`overlap_inflation`	up to 3.98-fold multi-PGS and 14.73-fold single-score inflation	Demonstrates overlap danger under a known covariance construction.
`real_meta_rules`	direct expected-R^2 proxy 0.17729 versus decorrelated 0.002261	Shows severe proxy sensitivity in one regime-C CAD diagnostic; cannot rank methods externally.

7 Unit tests and numerical contracts

The complete automated suite passed 400 of 400 tests in the environment of Appendix A. The contracts that carry the model, rather than the plumbing, are these.

Algebra and scale. Standardisation and back-transformation, coordinate updates, Gram ranges, projection identity and idempotence, and invariance to a positive rescaling of a score.

Fitting contracts. Shape and scale validation, tuning separated from assessment, frozen weights, convergence logs, deterministic seeds, and baseline fallback. Gaussian unpenalised columns match an explicit Frisch–Waugh–Lovell reference.

Genomic data handling. Allele orientation, ambiguous variants, duplicate handling, LD/GWAS scale agreement, and missing-data behaviour.

Evaluation. Continuous and binary metrics, the Lee transformation, and intervals that remain conditional on the supplied scores.

These tests pin the implementation to the mathematics in §3–§5. They do not pin it to a scientific claim about real GWAS.

Table 6: Automated verification layers and their interpretation.

Layer	Representative assertions	Result
Algebra and numerics	Standardisation and back-transformation, coordinate updates, Gram ranges, projection identity and idempotence, and score-rescaling invariance	Passed in the full test suite
Fitting contracts	Shape and scale validation, tuning separation, frozen weights, convergence logs, deterministic seeds, and baseline fallback	Passed in the full test suite
Genomic data handling	Allele orientation, ambiguous variants, duplicate handling, LD/GWAS scale agreement, and missing-data behaviour	Passed in the full test suite
Evaluation and documentation	Continuous and binary metrics, liability conversion, cited resources, benchmark result consistency, and example paths	Passed in the full test suite
Distributions	Expected source and wheel contents, installed import, and command-line smoke checks	Passed in the package build checks

8 Interpretation limits

1. **No causal interpretation.** Coefficients optimise prediction under correlation and penalisation. They do not estimate genetic correlations, mediation, or trait-specific causal contributions.

2. **No automatic overlap cure.** Cross-validation inside a target cohort cannot remove discovery contamination shared by every fold. Bivariate LDSC or named-cohort metadata may flag overlap under favourable conditions. Exclusion by design is stronger.

3. **No ancestry model.** Fitting weights in a target cohort does not repair discovery-score or LD-reference non-portability [19]. Absolute risk and score means are particularly unsafe across ancestries. Combining Catalog scores of many European-derived traits is not MIXPRS and does not substitute for a multi-population construction method [28].

4. **Direction matters.** Squared correlation hides sign reversal, and the training-free expected-R^2 routes assume consistent positive trait orientation.

5. **No clinical calibration.** A high ranking metric is not a calibrated probability. Calibration intercept and slope, subgroup performance, decision utility, and an externally justified prevalence are separate requirements.

6. **Conditional uncertainty.** The package bootstrap conditions on fixed component scores and their discovered weights. It does not propagate component-GWAS, LD-reference, or

architecture-estimation uncertainty. Incremental R^2 is truncated at zero, so a null score produces a bootstrap interval whose lower end is 0 by construction.

9 Conclusion

The central idea is ordinary. Move a difficult multi-trait genomic problem into a smaller space of component scores, then choose weights using the information that actually exists. The selection-index derivation says when gains are possible. The individual and summary Gaussian objectives say how weights are learned. The same-trait rules are transparent constructions whose accuracy assumptions must be checked externally.

That is not the MIXPRS problem. MIXPRS ensembles two multi-population methods of one trait after data fission [28]. This package stacks already-built scores, typically of many traits, or combines same-trait GWAS by metadata. The shared summary-statistic quadratic and PUMAS-style split [24, 25] do not identify the estimands.

The choice among routes is an information problem, not a universal ranking. Individual fitting is appropriate when a target phenotype and honest resampling are available. Summary fitting preserves the same score-space structure when GWAS cross-moments and LD are available. Training-free rules remain hypotheses until an untouched outcome is scored. For a same-trait method-by-ancestry ensemble, use MIXPRS rather than relabelling this package.

The committed tests and benchmarks demonstrate scale handling, numerical equivalence, controlled prediction gains, null behaviour, and sensitivity to LD-reference quality and discovery overlap. They explain how the software behaves under known conditions. They deliberately stop short of claiming clinical validity, portability, or calibrated risk.

A Reproducibility

Quantitative statements in §6 are generated from committed CSVs by `generate_evidence.py`. The input SHA-256 is the freshness authority; the source revision is generation context.

Package	`multipgs` 0.3.3
Source revision	`ca16e45e0cfb`
Input digest	`ea5ec47cac512410750d406c21369a7692426fc17c498e80d8dcc41a3f305274`
Tests	400 passed
Runtime	Python 3.10.20, NumPy 1.26.4, scikit-learn 1.7.2, pytest 9.1.1, ldpred3 0.4.7

The named artifacts are `fit_accuracy`, `null_gate`, `sumstat_vs_individual`, `sumstat_calibration`, `real_ld_simulation`, `overlap_inflation`, `meta_rules`, and `real_meta_rules`, each with a checked summary and a provenance file under `benchmarks/results/`.

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